Higher Colleges of Technology
كليات التقنية العليا

Computer and Information Science

Project

Course	CIS2303- Systems Analysis and Design		
Assessment Method	Group Project		
Date of Assessment	Week 15	**Deadline(s)**	
Maximum Mark	100	**Percentage of Final Grade**	25%

- The entire project/case study/poster is designed and developed by me (and my team members).
- The proper citation has been used when I (and my team members) used other sources.
- No part of this project has been designed, developed or written for me (and my team members) by a third party.
- I have a copy of this project in case the submitted copy is lost or damaged.
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CLOs	CLO1	CLO2	CLO3	CLO4	Total
Marks Allocated	**4**	**22**	**12**	**12**	**50**

Marks Obtained					

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CASE STUDY DESCRIPTION AND ASSUMPTIONS

Due to lack of a proper system, tasks take longer than they should. By creating a system for the nursery, admission will be developed quicker; take less time than it usually does, and will have a friendly interface for ease of use. The system will enhance proper and swift communication between the parent and teacher by the constant updates of the student's activities.

Admission managers will be able to view requests sent by parents and decide whether to reject or accept the kid based on specific criteria. Upon admission, parents will be able to track their kids' daily activities, and meal reports. The teachers will be the ones who will constantly update the system with the students' daily activities and meal reports. The system will notify the driver if the kid requires transportation and provide the driver with the student details (name and address).

ADOPTED METHODOLOGY

Our adopted methodology is Agile due to lack of a proper system and the tediously long task duration. Some of the requirements are not set and agile methodology helps us iterate and go through the steps cycle by cycle to refine and give us a satisfactory outcome of enhancing communication between the management and the parents.

FEASIBILITY STUDY

We need to conduct feasibility to help us determine whether or not the project will be conducted and finished successfully and measure how stable the development will be to the company.

TECHNICAL FEASIBILITY

To efficiently manage data storage and system needs, the Nursery Management System (NMS) will need a strong technical foundation. This comprises solid network infrastructure to provide constant performance and effortless data transfer, as well as powerful servers for effective workload control. User data, registration information, admission requests, and daily activity reports must all be kept in a secure database (Attana, 2024). Furthermore, cloud services like AWS, Azure, or Google Cloud can offer cost-effectiveness, scalability, and dependability, making it simple to increase processing and storage capacity as the system expands (Singhania, 2024). An additional degree of data safety is offered by cloud systems, which also offer strong backup and recovery options.

SCHEDULING FEASIBILITY

A meticulously planned methodology with iterative development phases to gradually offer critical functionality is necessary when creating a Nursery Management System (NMS) within an Agile framework. This approach gives the freedom to adjust to changing needs, take stakeholder input into account, and carry out continuous testing at every stage (Milne, 2021). A system that serves a variety of user groups, such as admission managers, parents, teachers, and drivers, must be able to adapt in order to efficiently meet their demands as the system evolves. Agile is perfect for developing a user-centered platform like the NMS because of its emphasis on responsiveness and collaboration (Lloyd, 2024).

ECONOMIC FEASIBILITY

A Nursery Management System (NMS) can be developed for between $120,000 and $220,000, which includes several necessary components. The estimated cost of software development, including database setup, front-end and back-end work, and hosting is between $20,000 and $50,000 each year. At $87,000 to $145,000, personnel costs—which comprise of developers, project managers, and testing support—are anticipated to be the biggest outlay. Workstation and tablet hardware may cost between $3,000 and $8,000. The annual cost of staff training and continuing maintenance is anticipated to be between $6,000 and $13,000. A complete system with development, support, and maintenance is guaranteed by this budget. (Dinnie, 2021)

Table 1

Component	Price
Hosting	$20,000 - $50,000
Personnel	$87,000 - $145,000
Workstation and hardware	$3,000 - $8,000
Staff training and maintenance	$6,000 - $13,000
Total	$120,000 - $220,000

OPERATIONAL FEASIBILITY

In order to ensure seamless system operation and adoption, the Nursery Management System's (NMS) operational viability requires effective workflows for all stakeholders. Automated notifications for new admission requests will let admission managers assess and make decisions quickly based on eligibility and available seats. Parents will have access to an easy-to-use portal where they can make admission requests, set up appointments, and get progress reports on their child's daily activities and meals, which are posted by teachers. In addition to setting up available times for parent appointments, teachers may effectively manage and upload student reports. Lastly, in order to streamline communication and cut down on delays, drivers will be automatically informed if they are assigned to transport a child along with the relevant information. All things considered, this technology improves operational efficiency by facilitating smooth communication and real-time updates among all stakeholders.

LEGAL FEASIBILITY

The platform must give data privacy, child protection, and digital security top priority in order to lawfully create a Nursery Management System (NMS). This entails abiding by laws such as FERPA (Branagan, 2021), which protects student record confidentiality, COPPA, which protects children's information, and GDPR, which governs data management (Jackson, 2023). Additionally, the system should have safeguards for minors, such limiting access to their data and requiring background checks for employees. HIPAA compliance is required if health data is involved. The platform must also be accessible to individuals with impairments and safe against data breaches. System restrictions and responsibilities will be outlined in explicit terms of use and service agreements. To make sure all standards are fulfilled, it is advised to speak with a legal professional. (Stephens, 2018)

REQUIREMENTS GATHERING AND TO-BE SYSTEM PERIMETER

REQUIREMENTS GATHERING TECHNIQUES

Our requirement gathering technique was a survey. We asked several questions about the system and how many features are provided and if the functionality feed the needs. We asked if they were satisfied with the overall performance of the system. We asked if it’s easy to navigate and what suggestions and feedback they have to say.

SURVEYS

Our outcomes were fairly positive most of the participants were very satisfied with the features and functionality of the new system. Some were somewhat satisfied with the performance, and some were normal about the performance, nothing extraordinary. The navigation of the system was said to be somewhat easy by 64% of our responses. We had a bunch of good suggestions and feedback that we will consider for future development like tighter security and add an easier interface as some of the participants were of old age.

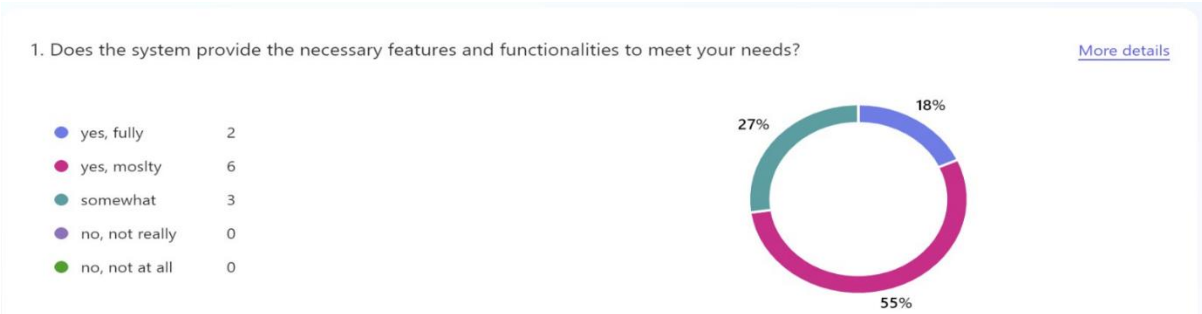

Figure 1

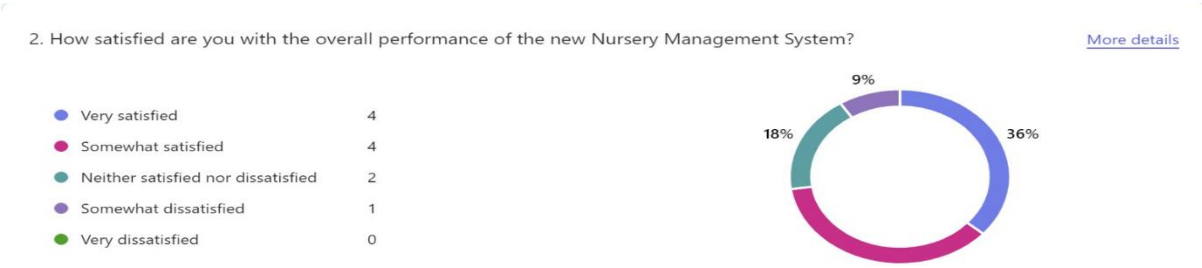

Figure 2

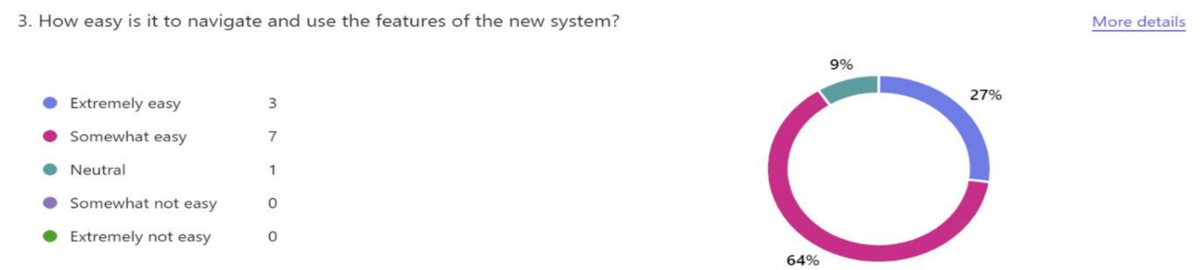

Figure 3

Figure 4

ACTIVITY DIAGRAM AND TO-BE SYSTEM PERIMETER

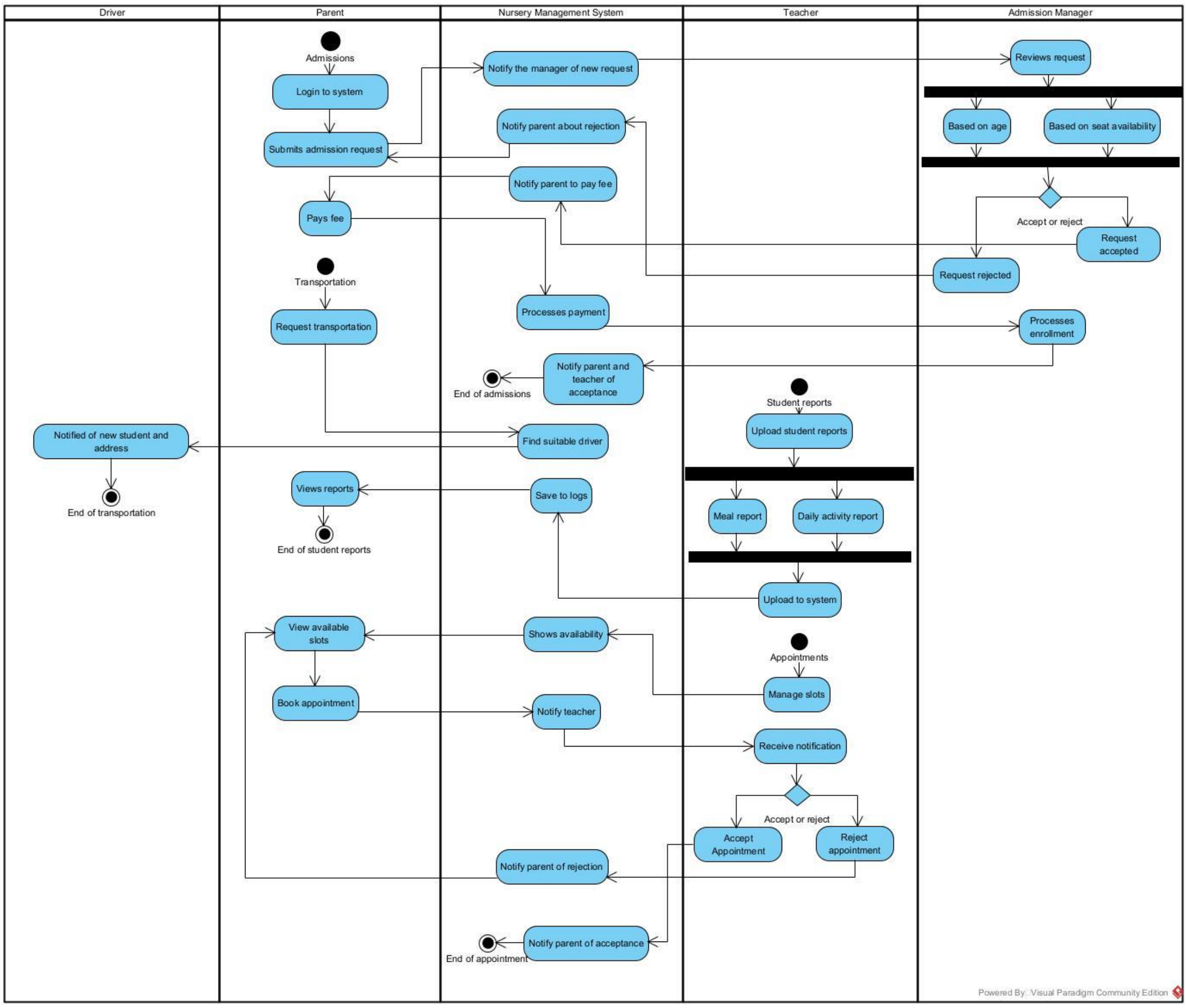

Figure 1

USER REQUIREMENTS SPECIFICATION

FUNCTIONAL REQUIREMENTS

For both parents and employees, the Nursery Management System will simplify necessary tasks. Admission supervisors evaluate applications according to age and seat availability, and parents can conveniently submit online admission requests, including any transportation requirements for their kids. Notifications are sent to parents and instructors upon acceptance, and drivers are promptly informed of any necessary transportation arrangements. Additionally, fee payments have been made simpler, enabling parents to make payments after admission is authorized. Teachers can post daily activity and meal reports for each kid, and parents can arrange meetings with teachers within available time slots for continued participation. Furthermore, the technology notifies parents and drivers when youngsters require transportation by allocating it automatically. For all those working in the nursery, this one-stop system guarantees efficient management and communication.

NON-FUNCTIONAL REQUIREMENTS

By making sure that private information, including student records and payment information, is safely maintained and only available to authorized users, the Nursery Management System will put security first. Additionally, it will be built to perform well, handling admissions requests fast and providing alerts on time. An intuitive interface that serves all users—parents, teachers, admission managers, and drivers—is essential for usability. While scalability will enable the system to accommodate an increasing number of kids and personnel as the nursery grows, reliability is crucial because the system should operate with little downtime. Lastly, the system will adhere to all local childcare standards and kid data protection laws, guaranteeing the ethical and secure handling of private data.

SPRINT 1 OR ITERATION 1

SYSTEM SEQUENCE DIAGRAM

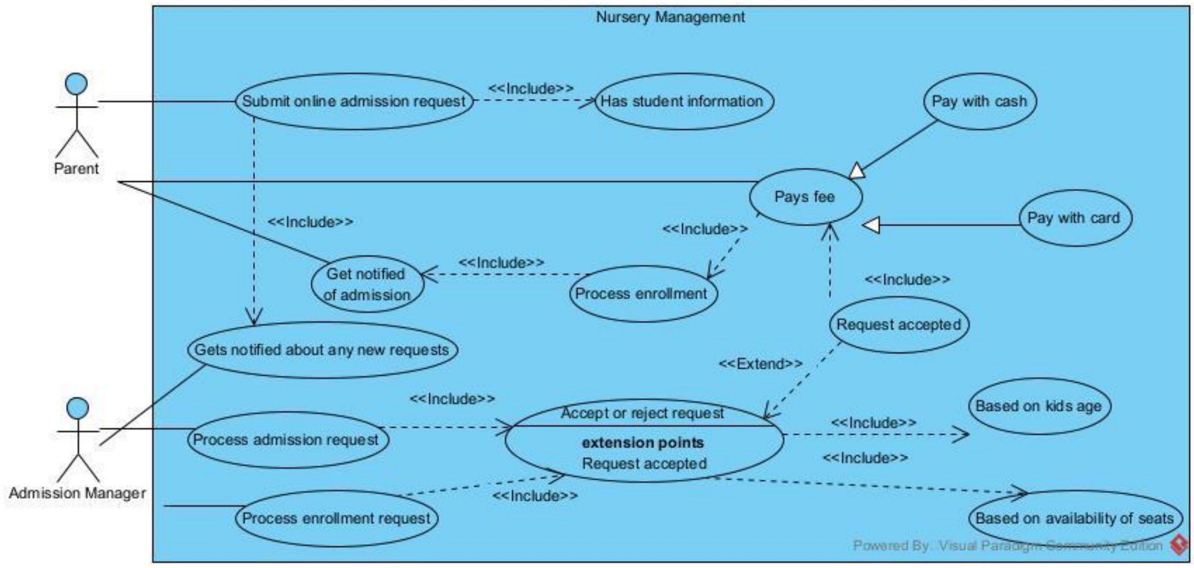

Figure 2

Table 2

Use Case	Request enrollment
Primary actors	Parent, Admission Manager

Summary	The use case describes the process by which a parent submits admission request (includes student information) and the Nursery Management System directs that request to an admission manager that then reviews the request and accepts or rejects it (based on kids age and their availability of seats) and once accepted, the parent pays the fee and they’re notified of acceptance.
Normal flow	• Parent submits online request that includes student information • Nursery management system directs that request to admission manager • Admission manager processes admission request • Admission manager accepts or rejects request based on kids age and availability of seats • Admission manager sends request to parent to make payment • Parent pays the fee either with cash or card • System accepts payment • Parent notified of successful enrollment
Alternative flow	Alternative flow (1): Request rejected • Admission manager rejects request • System directs invalid information (whether it’s based on kids age or availability of seats) to parent • Parent tries again next year Alternative flow (2): System rejects payment • System rejects payment • Parent tries again with another payment method

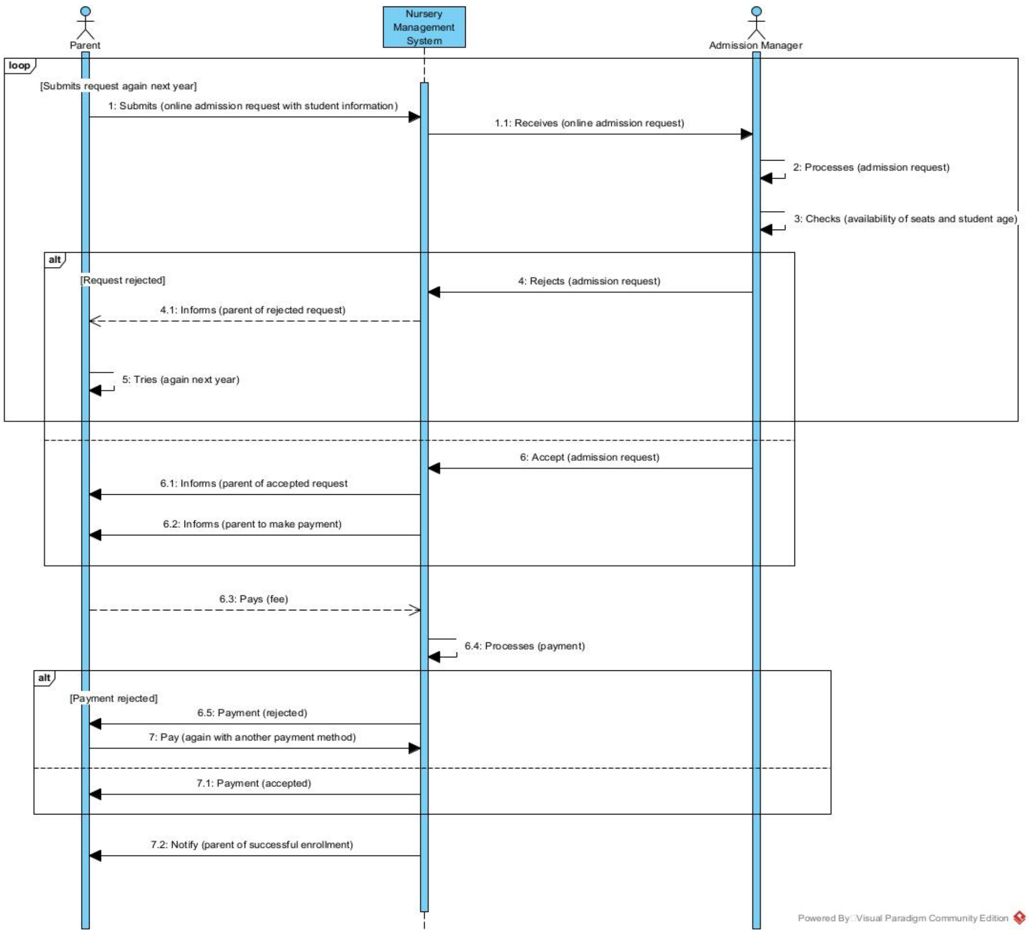

Figure 3

FRAGMENT OF THE DOMAIN CLASS DIAGRAM

Figure 4

SPRINT 2 OR ITERATION 2

SYSTEM SEQUENCE DIAGRAM

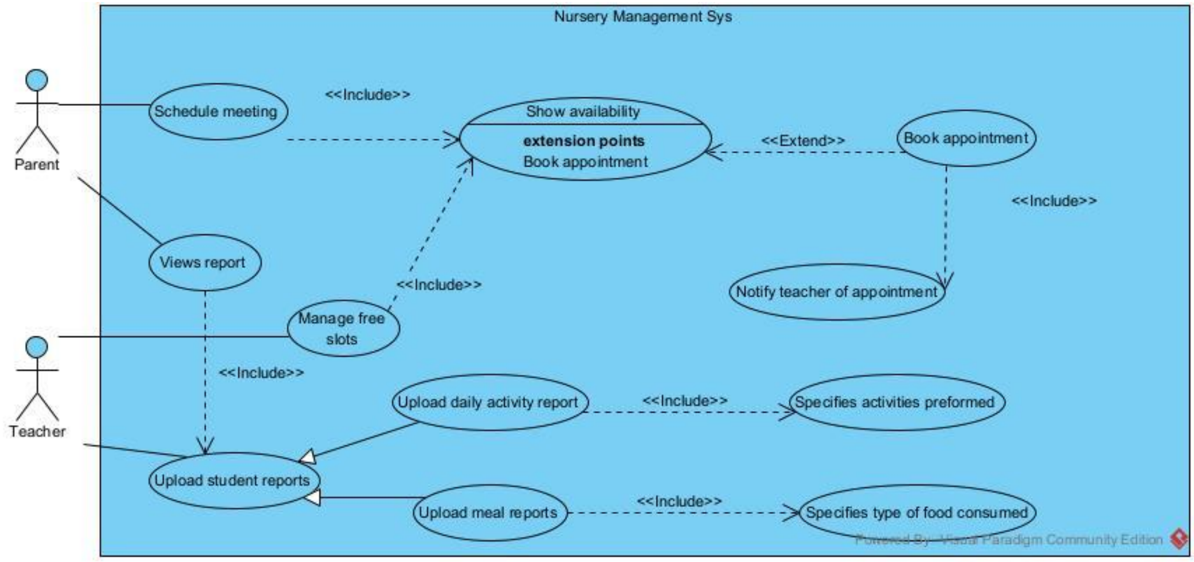

Figure 5

Table 3

Use case	Student report and scheduling meetings
Primary actors	Parent, Teacher
Summary	The use case describes the process by which a teacher uploads student reports (meal and activity reports) and a parent can view them. The parent can book appointment with teacher and the system shows free slots in the teacher's schedule. Upon booking an appointment, the teacher is notified of an appointment.
Normal flow	• Teacher uploads student meal reports and activity reports to system • Parent views reports from system • Teacher manages free slots • System updates schedule and shows availability • Parent views availability • Parent books appointment with teacher • System informs teacher of appointment
Alternative flow	Alternative flow (1): Parent tries to book on a booked slot • Parent books appointment • System shows invalid error about unavailability • Parent tries again with another slot

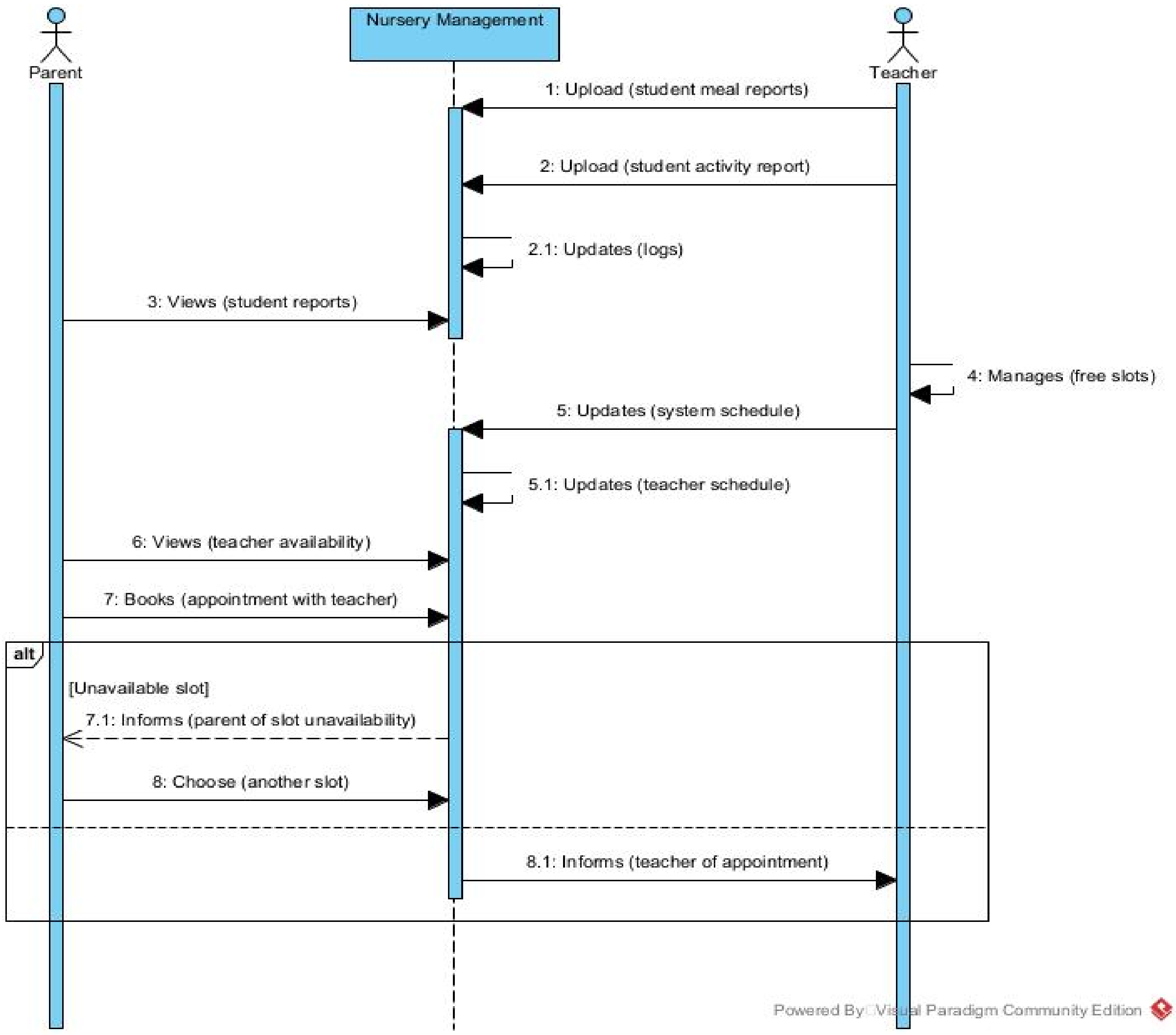

Figure 6

FRAGMENT OF THE DOMAIN CLASS DIAGRAM

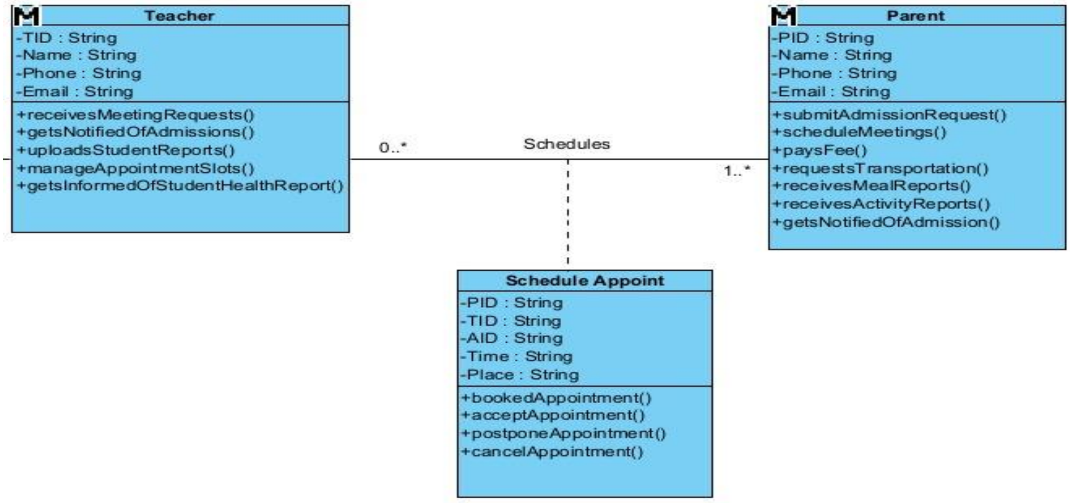

Figure 7

SPRINT N OR ITERATION N

SYSTEM SEQUENCE DIAGRAM

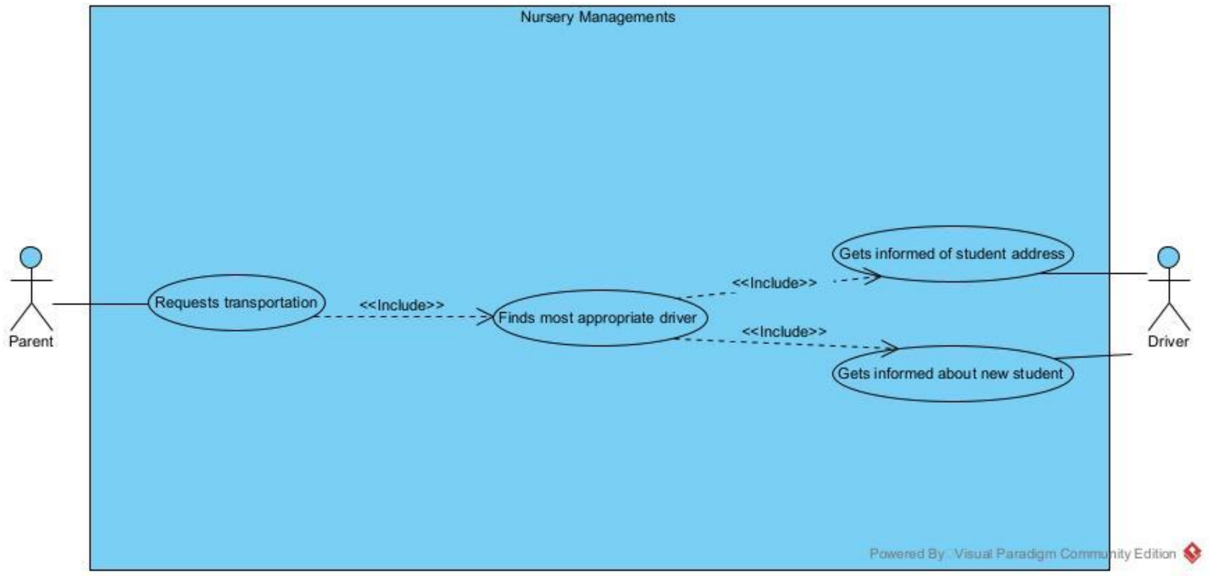

Figure 8

Table 4

Use Case	Request transportation
Primary	Parent
Secondary	Driver
Summary	The use case describes the process by which a parent requests transportation for their kid and the system finds the most appropriate driver that is then notified of student and address.
Normal flow	• Parent requests transportation

	• System finds the most appropriate driver • System informs driver of new student and address • Driver memorizes address and student
Alternative	Alternative flow (1): System fails to find appropriate driver • System unable to find driver • System informs teacher of lack of transportation • System makes special request • Parent tries again

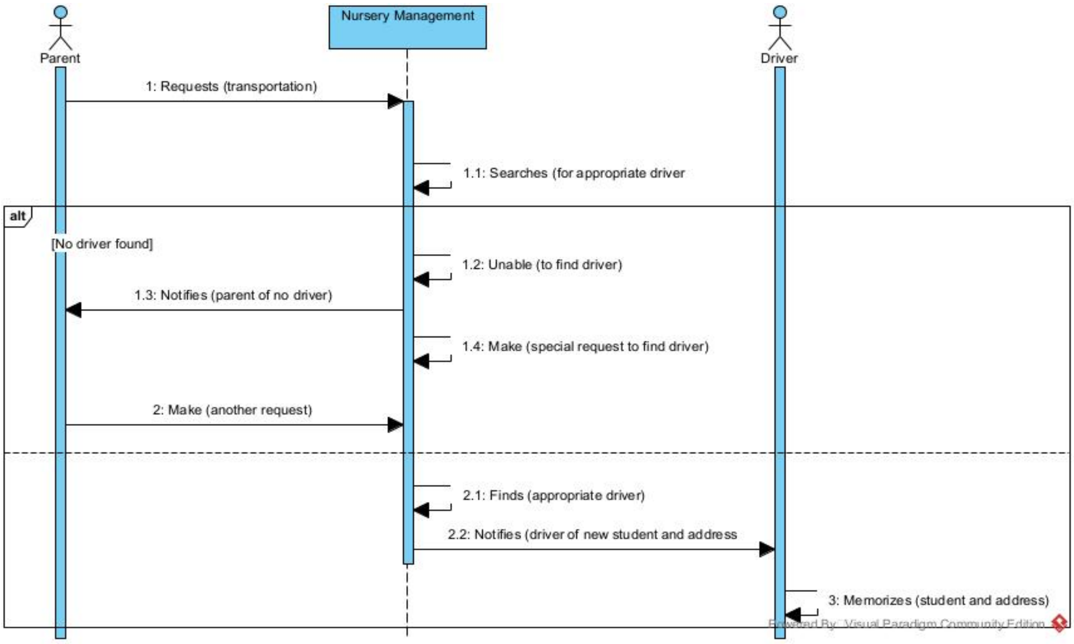

Figure 9

FRAGMENT OF THE DOMAIN CLASS DIAGRAM

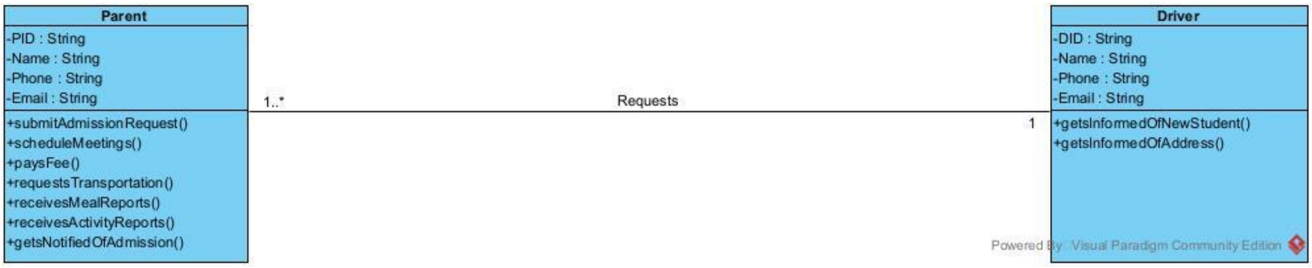

Figure 10

DOMAIN CLASS DIAGRAM

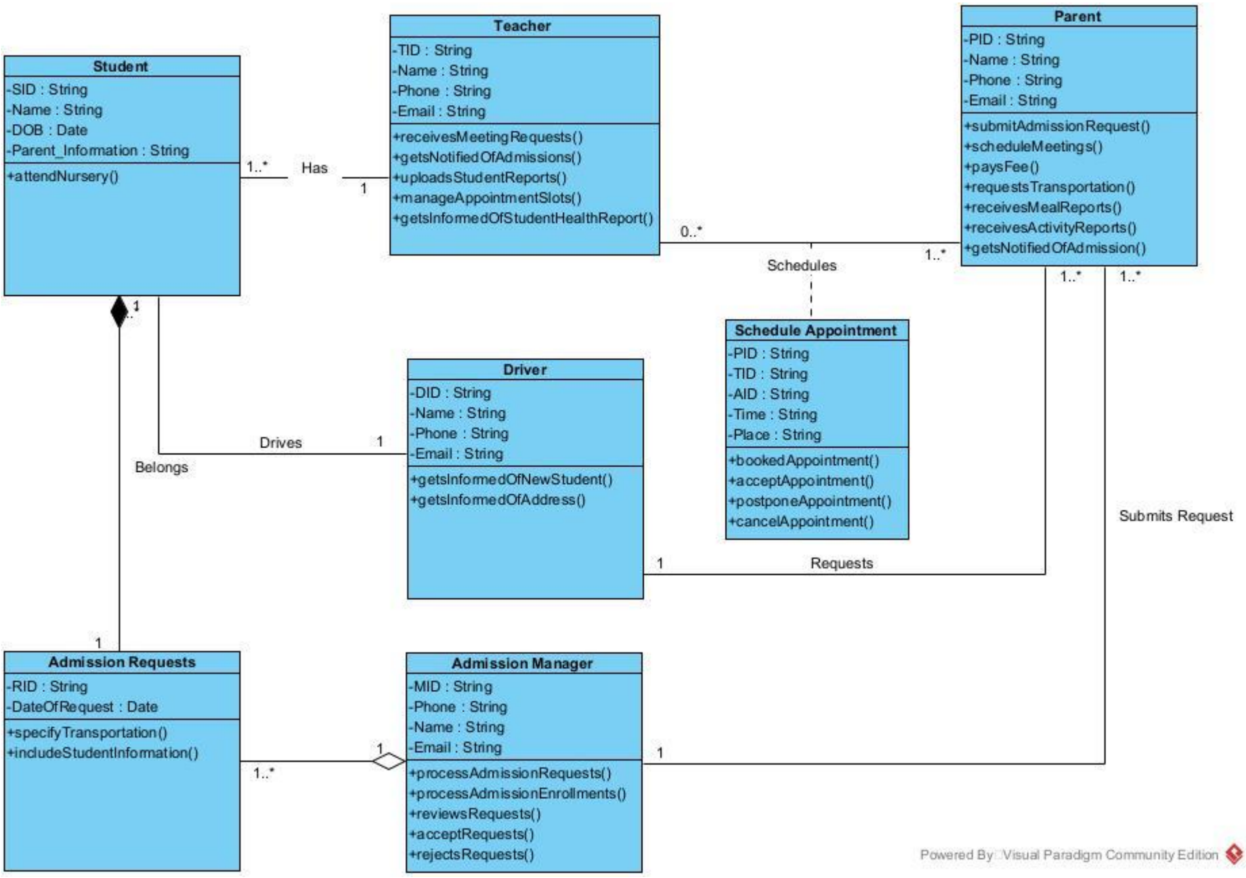

Figure 11

The class diagram explains how the system all works together. A student has one admission request that is reviewed by the admission manager. A teacher takes care of many students and a student has one teacher. Parents of student can schedule appointments with teacher in their free slots, teachers can accept or reject these appointment requests. A parent requests a driver for transportation, and a driver drives around many student, but a student has one driver.

TOWARDS DEFINING THE SOFTWARE ARCHITCTURE

ARCHITECTURAL STYLE

We chose Model-View-Controller (MVC) architectural design. The distinct division of responsibilities, which facilitates system maintenance and adaptation is the reason we selected it for our still growing system. The Model, View, and Controller are the three separate parts of the application that MVC separates. Because of this division, we may add new features or make upgrades without interfering with other system components. We can easily include additional features in this architecture as the system expands. For instance, we may alter the graphical user interfaces for admission managers, parents, and teachers on the View layer without changing the logic or data underneath. Additionally, each layer's independence guarantees that the system stays stable, flexible, and simpler to debug because each part works independently while contributing to the overall functionality of the system.

HIGH-LEVEL ARCHITECTURAL MODEL

To improve clarity, we started by breaking the Model up into two separate parts (see figure 12). We developed distinct packages for the enrollment process, and appointment scheduling. In order to ensure clear and effective interactions inside the system, each component is made to work in unison with the others while avoiding circular dependencies. After these packages were created, we incorporated them into our MVC architecture's Model layer.

We used a graphical user interface (GUI) in the View layer to efficiently display information to users (see figure 13). Each component is connected via dependencies, allowing smooth interaction between the Model and View levels. Because of the clear framework this structured arrangement offers within the MVC architecture, the system is simple to maintain, modify, and grow as needed.

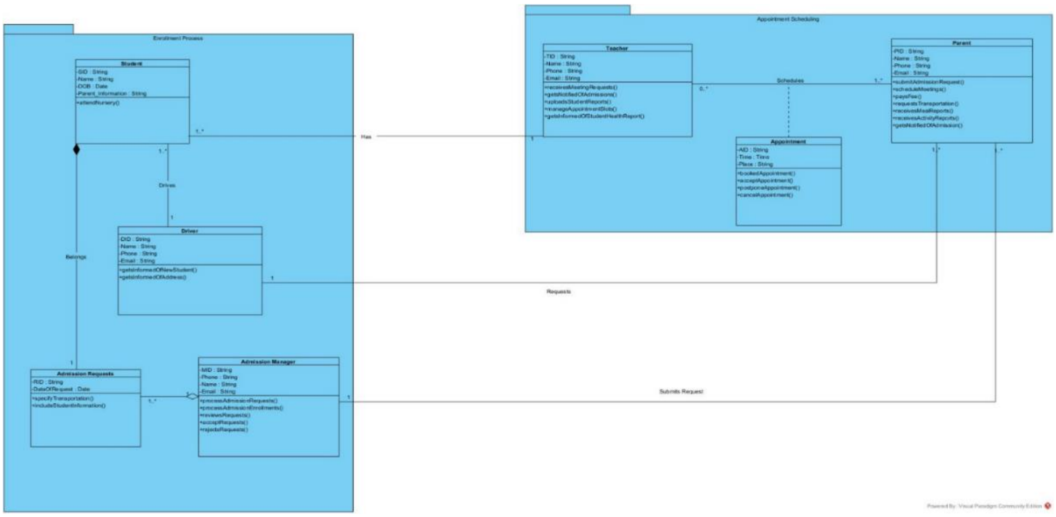

Figure 12

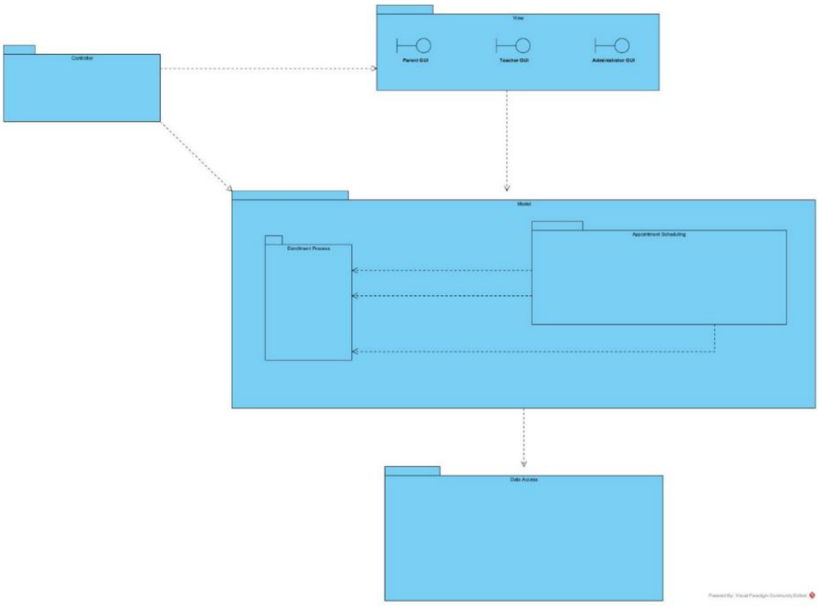

Figure 13

CONCLUSION

The development of the nursery management system is essential to help facilitate needs within the system and the people who use the system. With new features to enhance communication between all parties (teachers, parents, administrators, and drivers), the system addresses the inefficiencies caused by manual processes and delays. By implementing agile methodology, we were able to help the system take half the time it usually took and have an easier user interface for older people who will be using the system on a daily basis.

We took into consideration all aspects of feasibility study to make sure this system would have a successful outcome. The system guarantees data security and scalability to meet growing user needs and future expansion. The operational procedures guarantee that all parties involved can carry out their responsibilities efficiently and communicate with one another without any problems. The legal aspect and laws that protect children and ensure that private information is kept confidential. Finally, we took a realistic approach with the budget and their constraints, implementing it in our design.

In today's digital environment, we believe it is not only advantageous but also essential to invest in a strong Nursery Management System. A nursery management system is intended to reduce manual procedures, boost output, and enhance communication between all stakeholders. By putting such a system in place, operational effectiveness, data security, and adherence to pertinent laws will be guaranteed. The Agile methodology is also perfect for creating a user-centered platform that can expand over time because it fits in well with changing user needs.

Figure 17

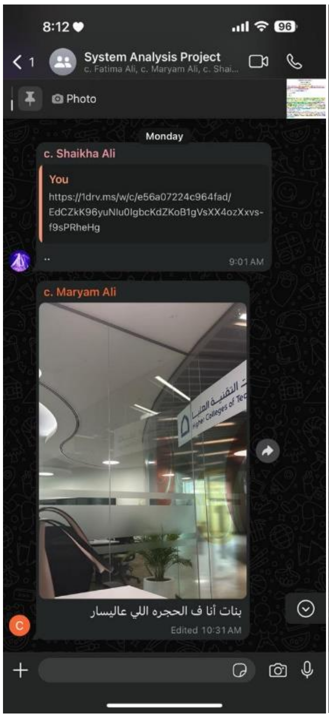

Figure 16

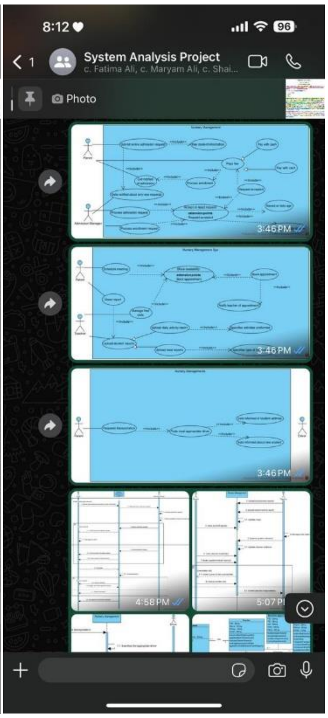

Figure 15

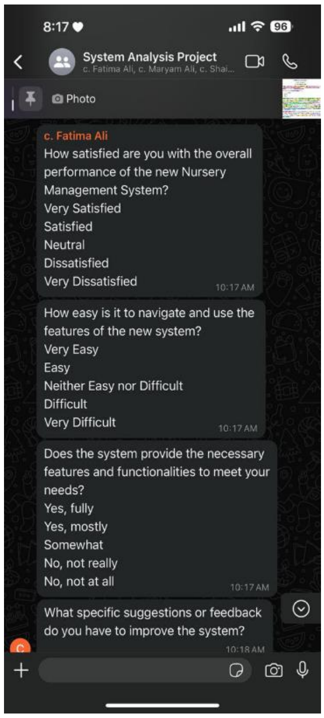

Figure 14

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